

Model quality in football: Quantifying the quality of an Expected Threat model

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Abstract

The recent growth in data availability in football has increased the risk of incorrect use of data-driven models, making guidelines on their validation and application necessary. The Expected Threat (xT) model is an accessible option for football organizations that start building in-house methods, yet little is known about how to assess its quality. The aim of this study is twofold: to examine how the model error depends on the number of game states and the number of training points, and to translate these results into guidelines for constructing and applying the model. Using the Markov chain underlying the model, we perform theoretical analyses and simulations to study the model error. These show that the model error is approximately log-normally distributed for a specified number of training points and game states. Additionally, we combine the simulations with expert consultation to establish the model error beyond which player evaluations based on the Expected Threat model become unreliable for scouting applications. From this, we derive rules of thumb to ensure the quality of an Expected Threat model before application, and we illustrate through an example how a validated model can be applied in practice. Because the approach generalizes to Expected Possession Value models, this paper illustrates a framework to systematically quantify model quality, despite the ground truth being unobservable in football analytics.

Keywords: Model error quantification, Possession Value Model, Markov chains, Concentration inequalities, Data-driven decision making, Football scouting

1 Introduction

Although the availability of data in (association) football has grown substantially over the last decade, there are few clubs that have in-house models to improve decision-making in their scouting or match analysis processes [1]. This, despite the potential impact ranging from improved talent identification to supporting tactical analyses [2–4]. For football clubs starting to build in-house models, accessible models can be more easily integrated into their processes. However, the use of easily accessible methods poses the risk of incorrect use of data techniques [5]. Post hoc argumentation and drawing conclusions that might be based on insignificant differences can be possible pitfalls. Consequently, it is important that these models are accompanied by guidelines on their validation and how and when to use them [1, 5].

A data-driven model that has generally been embraced by practitioners at football organizations is the Expected Goal model, often shortened as xG [6–8]. It models the probability of a shot being scored given a specific in-game situation. This can be used to evaluate shooting performance, goalkeeper performance, or scoreline analysis. The Expected Threat (xT/xThreat) model [9] generalizes the xG model. The Expected Threat describes the probability of scoring a goal in the same possession chain given the current game state. As the game state is defined by the position of the ball, it models the progression of the ball during a possession chain. The Expected Threat model makes it possible to evaluate the quality of general open-play actions, whereas xG only considers actions related to shots. These action evaluations provide information for opponent analysis or player scouting. The framework uses procedures that are intuitive to practitioners [10] and can be visualized in its traditional low-dimensional setting as illustrated by Singh [11]. This makes the Expected Threat model very accessible for football clubs starting to incorporate data into their decision processes.

Despite its accessibility, little is known about the quality of the model’s estimations. Differences between model and a game of football in reality arise both from modeling assumptions and from the model error. In this paper, we focus on studying issues related to the latter. More specifically, it is unclear how the number of different game states, indicative of the model complexity, and the size of the training set influence the errors of the threat values estimated by the model. Consequently, practitioners face challenges during model construction, application, and interpretation of the results.

The aim of this paper is twofold: (1) to obtain insights into the dependence of the error of the Expected Threat model on the number of game states and the number of training points, and (2) to provide guidelines for practitioners to build and apply an Expected Threat model. First, we give a mathematical description of the model. We then use this to find theoretical results on the error of the model, which provide a rough error bound and insights into the model behavior. The error of the model is studied in more detail via simulations, which give an approximate distribution of the model error for a specified training sample size and number of game states. Additionally, we perform simulations to find what size of the model error is acceptable for applications in player scouting. Finally, we use these results to provide guidelines to obtain a validated model and we illustrate how such a model can be applied to compare player performances at the 2020 Euros.

2 Expected Threat model

2.1 Theoretical framework

We first discuss the traditional Expected Threat model to establish the foundation of the work in this paper. The Expected Threat, $xT(s)$, is the probability that a goal is scored in the current possession chain in football, given the current game state s . The concept was first introduced by Rudd [9] and later popularized in the blog post by Singh [11]. In the traditional formulation, the game state is defined by the location of the ball during a possession. The possession chains in the game are described by a Markov chain. Then, the probability of scoring given the current state is a well-defined and can be estimated.

Because the aim in football is to score goals, a good pass or dribble increases the likelihood of scoring. The quality of an on-the-ball action is therefore described in (1) as the difference in Expected Threat before and after an action. Here, the action $a_i = (s_{i-1}, s_i)$ moves the game from state s_{i-1} to s_i .

$$\Delta xT(a_i) = xT(s_i) - xT(s_{i-1}), \quad \text{where } a_i = (s_{i-1}, s_i) \quad (1)$$

The Expected Threat model assumes that football adheres to the Markov assumption: distribution of the future states in a football game only depends on the current state and not on the past. When the game is in state s , a goal could be scored if the player with the ball takes a shot, and the shot is successfully scored. A goal can also be scored by transitioning to another game state s' and scoring from that position. The probability of scoring from that position is again the Expected Threat at state s' : $xT(s')$. Consequently, the Expected Threat model should satisfy the equation as given in (2).

$$xT(s) = P(\text{shot}|s) \cdot xG(s) + \sum_{s' \in S} T_{s,s'} \cdot xT(s'), \quad (2)$$

where $xG(s)$ is the probability of scoring a goal given a shot from state s , also known as the extensively-studied Expected Goal (xG) value, S is the set of all game states and $T_{s,s'}$ is the probability of the game transitioning from state s to state s' .

When the set of game states S is of finite size M , the Expected Threat model allows for an alternative notation with vectors and matrices. To do this, the Expected Threat is denoted as $xT \in [0, 1]^M$. The probability of scoring a goal can be contained in one vector $g \in [0, 1]^M$, which is element-wise defined as $g_s = P(\text{shot}|s) \cdot xG(s)$. The transition matrix $T \in [0, 1]^{M \times M}$ gives the probabilities from each in-game state to each other in-game state. The defining equation can then be reformulated as

$$xT = g + T \cdot xT. \quad (3)$$

2.2 Estimation

In practice, the quantities in (3) are unknown and have to be estimated. In recent years, the quantity of collected data in football has increased, resulting in often proprietary data sets [5, 12, 13]. With historical data of in-game actions, the values of $P(\text{shot}|s)$,

$xG(s)$, $T_{s,s'}$, or the vector g and matrix T can be estimated. This is generally done by counting occurrences and taking the empirical mean.

The only unknowns in the model then remain the xT -values, which are solutions to a linear system of equations. This can be solved with various methods, but, a specific version of the Value Iteration algorithm [14] is used that is intuitive to coaches and players.

This Value Iteration algorithm calculates the probability of scoring within the next k actions. Scoring a goal within one action can only be done by directly shooting and scoring successfully. The initial value is therefore set as $P(\text{shot}|s) \cdot xG(s)$ or $xT^{(1)} = g$. Scoring within k actions is possible by scoring directly or moving the ball and then scoring a goal within $k - 1$ actions. This allows the iterative calculation of the probability of scoring within k actions, and this algorithm is summarized in Algorithm 1. The values of this algorithm converge to the correct values when $k \rightarrow \infty$, as proven in Proposition 1.

Algorithm 1 Vectorized Value Iteration algorithm for calculating xT values

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1: Input: Convergence threshold  $\varepsilon > 0$ 
2: Known values:  $g \in [0, 1]^M$ ,  $T \in [0, 1]^{M \times M}$ 
3: Unknown:  $xT \in [0, 1]^M$ 
4:  $xT^{(1)} \leftarrow g$ ,  $k \leftarrow 1$ 
5: while  $\|xT^{(k)} - xT^{(k-1)}\| > \varepsilon$  do
6:    $xT^{(k+1)} \leftarrow g + T \cdot xT^{(k)}$ 
7:    $k \leftarrow k + 1$ 
8: end while
9: return  $xT^{(k)}$ 

```

2.3 Definition of game state

An important aspect of the Expected Threat model is how the game states are defined. Traditionally, the field is discretized and divided into different rectangles. The rectangle in which the ball-possessing player is located during a possession chain gives the game state [11]. A common discretization of the pitch is a 16×12 grid. This grid roughly follows the ratio between dimensions of the football pitch, and it results in a model with $M = 16 \cdot 12 = 192$ game states. In this way, the discretization of the pitch in two dimensions determines the game states of the model.

The discretization grid is a tuning parameter of the Expected Threat model. A model with a finer grid is better able to distinguish between different in-game situations and could be considered favorable. On the other hand, a finer grid results in more probabilities to estimate as the sizes of g and T grow. With a fixed size of the data set, this decreases the number of data points that can be used to estimate a single probability and lowers the quality of the estimation. As a result, the choice of the grid is a model tuning parameter that strongly influences the model precision and quality of the estimates.

The traditional approach of defining game states solely by the ball-possession player’s positions does not capture the full game dynamics that influence scoring probability [10]. Van Arem and Bruinsma [15] extended the traditional model by adding two extra dimensions: an indicator for a high or low ball and the number of defenders between the player and the goal. Because this rapidly increases the number of game states, they estimated the xG-values via an existing xG-model and used kernel density estimation to prevent overfitting. Hassani et al. [16] added the xG-value as an extra dimension to the Expected Threat model. Based on the expected behavior of football players, they assumed extra structure on the transition probabilities. In this way, the Expected Threat model can be extended to better capture game dynamics.

In a broader sense, the Expected Threat model is an Expected Possession Value (EPV) model. EPV models are often studied with continuous state spaces, and scoring probabilities are generally estimated through neural networks such as (convolutional) neural networks [17, 18], graph neural networks [19], or autoencoder-decoders [20] to estimate the probability of scoring. These models can be used to assess decision making [21–23] and assess defensive quality [24]. Hereby, they can provide extra insights for clubs with a dedicated data department.

However, these more complex approaches require tracking data, which is less widely available. This makes it harder to apply those models in scouting processes, where tracking data is less accessible. Additionally, the described methods perform smoothing or regularization to deal with the increased number of the game states, which complexifies the mathematical analysis of the model error. Therefore, this paper focuses on the traditional formulation of the Expected Threat model, although most of is applicable to general EPV models.

2.4 Quantifying the model error

The Expected Threat model aims to approximate the game of football as closely as possible. There are two factors that may cause a difference between model and reality. First, the model makes assumptions such as the Markov assumption and the definition of the game states. Players are influenced by the past, albeit slightly, and the threat on the field changes gradually instead of with few discrete steps. This causes a discrepancy due to the modeling assumptions.

Second, an error is made within the context of the assumed model. Estimating $\hat{g}$ and $\hat{T}$ and applying Algorithm 1 can introduce deviations from the true values. Because this error arises within the context of the assumed model, we refer to it as the model error. Both influence how closely the Expected Threat model describes reality. Although we acknowledge that a finer grid decreases the influence of the model assumptions, our main focus in this paper is to study the model error, since only little is known about this aspect.

To study the size of the model error, we assume that there exist true xT -values for each set of game states S of size M . These values are the underlying probabilities of scoring a goal in each defined game state. When training an Expected Threat model, these quantities are estimated by taking empirical means and applying Algorithm 1. The resulting model error is $xT - \widehat{xT}^{(k)} \in [0, 1]^M$, where xT represents the true value

and $\widehat{xT}^{(k)}$ the estimated value after k iterations. The size of this error can then be measured with any vector norm: $\|xT - \widehat{xT}^{(k)}\|$.

In the context of this paper, the ℓ_∞ -norm, defined as $\|xT - \widehat{xT}\|_\infty = \max_{s \in S} |xT(s) - \widehat{xT}(s)|$, is a suitable candidate for three reasons. First, this norm does not require rescaling to compare models among different number of game states M . Second, the probability of scoring a goal is close to 0 far away from the goal, and the error in estimating these xT-values is low in practice. By taking the maximal values, many small errors do not deflate the measured error when significant errors are made at critical states close to the goal. Finally, the ℓ_∞ -norm provides an intuitive way of measuring the error of the model as it describes the maximal estimation error for each single state. This gives practitioners a clear handle to interpret the margin of error when applying the model at football clubs. The size of the model error is therefore measured as $\|xT - \widehat{xT}\|_\infty$.

3 Theoretical results

The goal of training an Expected Threat model is to estimate the true underlying xT -values, but errors can occur through the estimation process. There are two parts of the process that can cause errors. First, the empirical estimation of the vector g and in-game transition matrix T causes an error. A second source is the numerical error created by Algorithm 1, which calculates the estimated xT -values from the estimated g and T . This can be summarized as

$$xT - \widehat{xT}^{(k)} = (xT - \widehat{xT}) + (\widehat{xT} - \widehat{xT}^{(k)}), \quad (4)$$

where xT denotes the vector with the true underlying values, $\widehat{xT}$ the statistical estimate based on a sample of size N and $\widehat{xT}^{(k)}$ the estimated values after k iterations through Algorithm 1.

3.1 Value Iteration algorithm

The term $\widehat{xT} - \widehat{xT}^{(k)}$ in (4) can be considered the numerical error caused by Algorithm 1 after k iterations. The following result gives a bound on this numerical error. It is formulated with the estimated quantities $\widehat{xT}$, $\widehat{xT}^{(k)}$, $\hat{g}$, and $\hat{T}$. The result also holds for the true underlying values of xT , $xT^{(k)}$, g , and T .

Proposition 1 *Consider an estimated Expected Threat model as a Markov chain. Assume that $\|\hat{T}\|_\infty < 1$. Then, the error of the Value Iteration algorithm of the Expected Threat model (Algorithm 1) after k iterations satisfies*

$$\|\widehat{xT} - \widehat{xT}^{(k)}\|_\infty \leq \frac{\|\hat{g}\|_\infty \|\hat{T}\|_\infty^k}{1 - \|\hat{T}\|_\infty}. \quad (5)$$

The assumptions of this proposition are easily satisfied in practice. The estimated $\hat{T}$ and $\hat{g}$ also define a Markov chain themselves. This Markov chain contains two implicit Markov states in addition to the game states: scoring a goal and loss of possession. When every state has a nonzero probability of either scoring a goal or losing possession, the assumption $\|\hat{T}\|_\infty < 1$ holds. If this is not satisfied in practice for a specific state, the Markov chain can be modified by dropping the state from the chain and redefining the transitions accordingly. Because the assumptions are easily satisfied, the numerical error is bounded in practice by the right hand side in (5).

As a result of Proposition 1, the error decreases geometrically at each iteration of Algorithm 1 with a factor $\|\hat{T}\|_\infty$. In practice, the transition matrix is sparse, and the algorithm can be implemented such that iterations are relatively cheap. Consequently, the number of iterations can generally be chosen such that the numerical error is negligible.

3.2 Statistical Error

Since the numerical error can be made arbitrarily small by performing sufficiently many iterations, the model error mainly arises from the statistical estimation of the quantities. The statistical error splits into the estimation error of the vector g and the matrix T . A probabilistic bound on these statistical errors can be obtained via concentration inequalities. These insights can then be combined to find one probabilistic bound on the size of the statistical error.

For this purpose, a scenario is considered where there is a dataset of N in-game actions, for instance, all actions of a season in a league. Let N_g denote the number of shots and let N_m be the number of actions that move the ball. Note that $N = N_g + N_m$ and that the proportion of shots is constant. This means $N_g \approx p_g N$ and $N_m \approx (1 - p_g)N$, where $p_g \approx 0.02$ is a constant that represents the proportion of shots in the dataset. Then, N , N_g , and N_m are of the same order of growth. For the purpose of this analysis, it is assumed that the shots and actions are uniformly distributed over all in-game states, i.e., each state has on average $\bar{N}_g = N_g/M$ shots and $\bar{N}_m = N_m/M$ actions that move the ball on the pitch.

3.2.1 Split of statistical error

The size of the statistical error can be split into a term representing the error in the goal probability and one representing the error in the transition probabilities. This is described in the following proposition.

Proposition 2 *Let xT , g , and T denote the quantities of the true Expected Threat model, and let $\hat{xT}$, $\hat{g}$, and $\hat{T}$ denote the quantities of the estimated Expected Threat model. If $\|T\|_\infty < 1$, then*

$$\|xT - \hat{xT}\|_\infty \leq \frac{\|g - \hat{g}\|_\infty + \|(T - \hat{T})\hat{xT}\|_\infty}{1 - \|T\|_\infty}. \quad (6)$$

The proposition shows that the statistical error consists of the error in the estimation of model parameters g and T with a rescaling factor. The error of the transition

probabilities is propagated to the statistical error, but only after being weighted by the estimated xT -values of the corresponding states. Consequently, errors in transition probabilities are most problematic if they happen in states with a high estimated xT -value.

The rescaling factor $\frac{1}{1-\|T\|_\infty}$ is large if $\|T\|_\infty \approx 1$. This happens if a game state has a very small probability of both losing ball possession and scoring a goal. In such situation, the bound will increase, because the xT -values are difficult to estimate for this Markov chain.

3.2.2 Concentration inequalities

The model parameters g and T are estimated via the empirical mean. Concentration inequalities can be used to derive bounds on the estimation error of both. This results in bounds in terms of the number of game states M and the size of the training data set N . Combined, they provide a bound on the statistical error and can give insights into the impact of estimation errors in different model parameters.

The goal-scoring vector, g , gives the probability of scoring given ball possession in game state s . For each state, it holds that $P(\text{goal}|s) = P(\text{goal}|\text{shot}, s) \cdot P(\text{shot}|s)$. The estimate $\hat{g}$ is often obtained by estimating the probabilities of a shot being taken, and the shot being successful with the empirical mean. These are then multiplied to find the value $\hat{g}_s$. Equivalently, one could estimate g by taking the empirical mean of all goals being scored from all visits to a state. The following proposition gives a bound on the error of such estimation.

Proposition 3 *Suppose that each entry of g is estimated by the empirical mean of $\bar{N}_g$ independent Bernoulli random variables with expectation g_s . Then, the bound*

$$\|g - \hat{g}\|_\infty < \sqrt{\frac{\log(2M/\alpha)}{2\bar{N}_g}} = M^{1/2} \sqrt{\frac{\log(2M/\alpha)}{2N}} \quad (7)$$

holds at least with probability $1 - \alpha$.

Proposition 3 assumes that the observed transitions leaving each state s are independent random variables. Note that this is required within each state only and independence between samples from different states is not assumed. Because the Markov assumption gives that different visits to a specific state are independent, this assumption holds for the Expected Threat model.

The transition matrix gives the probabilities of moving from one in-game state to another in-game state. These are estimated by taking the empirical mean for each individual transition probability. The following proposition gives a bound on the error of the estimated transition matrix.

Proposition 4 *Suppose that each entry of T is estimated by taking the empirical mean of $\bar{N}_T$ independent Bernoulli random variables with expectation $T_{i,j}$. Then, the bound*

$$\|T - \hat{T}\|_\infty < M \sqrt{\frac{\log(2M^2/\alpha)}{2\bar{N}_m}} = M^{3/2} \sqrt{\frac{\log(2M^2/\alpha)}{2N}} \quad (8)$$

holds with probability at least $1 - \alpha$.

Proposition 3 shows that the bound on the error in estimating g grows with order $M^{3/2} \sqrt{\log(M)}$. According to Proposition 4, the estimation error is bounded by a quantity growing with order $M^{1/2} \sqrt{\log(M)}$. The error when estimating the transition matrix will grow faster than the error for estimation of the goal-scoring probabilities for large values of M .

The dependence on N shows the same behavior for both model parameters. Both bounds decrease with an order of $1/\sqrt{N}$. As expected, the estimation of the model parameters is better for large values of N .

3.2.3 Model error

The derived results can be combined to obtain one bound on the error of the estimated Expected Threat values. The following theorem is a direct consequence of the derived propositions. As discussed in Section 3.1, the assumptions for this theorem are naturally satisfied during application.

Theorem 1 *Let xT , g , and T denote the quantities of the true Expected Threat model with M game states, and let $\widehat{xT}^{(k)}$, $\hat{g}$, and $\hat{T}$ denote the quantities of the estimated Expected Threat model with k iterations of Algorithm 1. Assume that $\|T\|_\infty < 1$ and $\|\hat{T}\|_\infty < 1$. Then, with probability $1 - \alpha$, the error of the estimated model is bounded by*

$$\begin{aligned} \|xT - \widehat{xT}^{(k)}\|_\infty &< \underbrace{\frac{1}{1 - \|T\|_\infty} \left(\overbrace{\frac{1}{\sqrt{p_g}} M^{1/2} \sqrt{\frac{\log(2M/\alpha)}{2N}}}^{\text{Error } \hat{g}} + \overbrace{\frac{1}{\sqrt{1-p_g}} M^{3/2} \sqrt{\frac{\log(2M^2/\alpha)}{2N}}}^{\text{Error } \hat{T}} \right)}_{\text{Statistical error}} \\ &+ \underbrace{\frac{\|\hat{g}\|_\infty \|\hat{T}\|_\infty^k}{1 - \|\hat{T}\|_\infty}}_{\text{Numerical error}} \end{aligned} \quad (9)$$

The two terms corresponding to the errors of $\hat{g}$ and $\hat{T}$ can be combined into one by taking the highest order terms. The numerical error is negligible in typical situations, as discussed in Section 3.1. Then, the last term of the bound can be ignored. The fact that $p_g \ll 1$ gives that $\frac{1}{\sqrt{1-p_g}} \approx 1$. These insights can be combined to find the following approximate bound with probability $1 - \alpha$.

$$\|xT - \widehat{xT}^{(k)}\|_\infty \lesssim \frac{p_g^{-1/2} + 1}{1 - \|T\|_\infty} M^{3/2} \sqrt{\frac{\log(2M^2/\alpha)}{2N}}. \quad (10)$$

This bound shows that the error is in terms of N and M at most of order $O(M^{3/2}\sqrt{\log(M)}/\sqrt{N})$ with high probability. It shows that the model error is expected to grow when the number of game states M is increased or the number of training data points N is decreased.

However, the errors of transition probabilities and the goal probabilities have different multiplication factors. Because p_g is small, the term with a factor $M^{1/2}\sqrt{\log(M)}$ term gets a higher multiplicative constant. For finite values of M , the error in goal-scoring probabilities $\|g - \hat{g}\|_\infty$ can be dominant instead of the error in the transition probabilities $\|T - \hat{T}\|_\infty$, which has a higher order.

The interpretation of the game of football also provides extra nuance. States with a low threat tend to be visited more often. These low-threat states often have a low probability of scoring directly and many data points, leading to good estimation of the values g . Similarly, the error in estimating T is weighted by $\widehat{xT}$. This means that low-threat game states contribute little to the model error. Because only a few game states are of high threat, most game states typically have small model errors.

Because many game states have small errors and the error in estimating $\hat{g}$ might be dominant for finite values of M and N , we hypothesize that the error of the Expected Threat model is lower in typical applications. The orders might grow less fast than the worst-case $O(M^{3/2}\sqrt{\log(M)}/\sqrt{N})$. Although the theoretical results provide insights into the behavior of the model, the typical behavior of the model error during application is unknown.

4 Simulations

Two types of simulations were performed that provided insights into the typical behavior of the Expected Threat model. First, we performed a simulation study to obtain an approximate distribution of the model error given realistic values of both the training sample size N and the number of game states M . Second, we combined expert consultation with simulations to find which model errors result in reliable player ratings for scouting purposes. These simulations provided information on the size of the model error and the influence of model error on the quality of player ratings. With this, we will formulate rules of thumb to constrain the error of the Expected Threat model during application in Section 5.

4.1 Methods

For a specific grid of size M , we trained an Expected Threat model on the openly available StatsBomb open data [25]. For the purpose of the analysis, these models were assumed to be the ground truth.

Based on the Markov chain in the model, a new data set of size N can be simulated by sampling different possession chains. A new model with the same grid can be trained, which estimates the ground truth model. As both the ground truth xT and the estimated $\widehat{xT}$ are known, the size of the model error can be calculated for the estimated model. The simulated model error is a bootstrap approximation of the model error. This process is visualized in Figure 1 and yields a situation where the error of

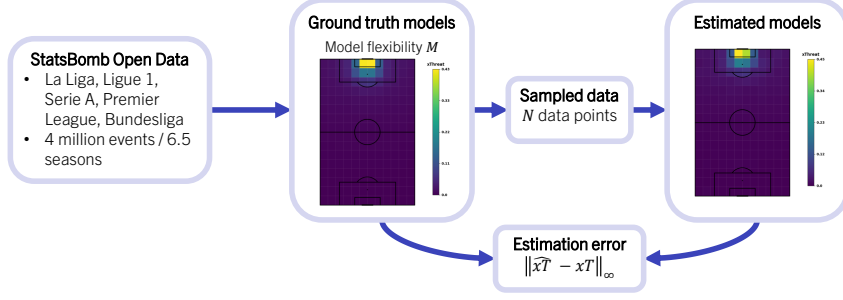


Fig. 1: A visualization of the simulation scheme used in the study.

the model can be assessed despite the general lack of ground truth in sports modeling [26].

To perform simulations, we used the openly available StatsBomb open data [25] to train the ground truth model. All available games from the Premier League, Ligue 1, Serie A, La Liga, and the Bundesliga were used. The events that did not describe passes, dribbles, errors, clearances, or shots were filtered out. This resulted in a dataset of approximately 4 million in-game events, or roughly 6.5 full seasons of a single league.

The code for the simulations are available on [GitHub](#). The computations were performed on the DelftBlue Supercomputer [27].

4.1.1 Distribution of the model error

Simulations were carried out to investigate the behavior of the model error in practice. The sampling process was repeated 1,000 times for different values of N and M . The combinations of the values of N are given in Table 1 and those of M with corresponding grid dimensions in Table 2. The grids were chosen to be of roughly the same proportions as the traditional 16×12 grid. The simulations resulted in 104,000 estimated models with different values of M and N , with their corresponding error quantities.

Some combinations of M and N resulted in errors too large for practice. To exclude these unusable combinations, a filter was applied $M^{3/2} \log(M) / \sqrt{N} \geq 445$. The value of 445 was empirically determined to filter out the situations with a mode of the model error between 0.8 and 1.0. The necessity of this filter was examined via the unfiltered samples. For each combination of M and N , kernel density estimates of the model were computed and compared.

For the resampled models, the model parameters were estimated in the process. The estimation error of these parameters could also be calculated. To find the importance of the errors for different model parameters, the relationship between the model error and the error in the model parameters were visualized through scatter plot, Pearson correlation coefficients, and histograms.

Because the size of the model error $\|xT - \widehat{xT}\|_\infty$ is the maximal absolute difference of all xT values, it can only attain positive values. This means that an assumed statistical model for this error needs to have positive support. From the theoretical analysis, it is known that the error becomes larger for increasing values of M and decreasing

Table 1: Sample sizes N used in the simulations to study the distribution of the model error.

Sample sizes N		
100,000	240,000	1,300,000
130,000	370,000	4,000,000
170,000	630,000	

Table 2: Grid sizes and number of game states M used in the simulations to study the distribution of the model error.

Grid dimensions: $m_x \times m_y$ (M)			
8×6	(48)	20×15	(300)
10×8	(80)	24×18	(432)
12×9	(108)	28×21	(588)
14×11	(154)	32×24	(768)
16×12	(192)	40×30	(1200)

values of N . As the error is positive, both the variance and mean can be expected to depend on N and M . Still, the hypothesis is that the influence of M and N on the model error is in reality of a lower order than the worst-case $O(M^{3/2}\sqrt{\log(M)}/\sqrt{N})$. A statistical model with a multiplicative error and different powers of M and N is desirable.

A parametric model with a lognormal distribution was assumed as follows.

$$\|xT - \widehat{xT}\|_\infty = CM^\alpha/(\sqrt{N})^\beta e^\varepsilon, \quad \text{where } \varepsilon \sim N(0, \sigma^2). \quad (11)$$

This random variable only attains positive values. The parameters describing the powers of N and M make it possible to scale both the mean and expected value of the error. Additionally, they allow for inference on the powers for N and M .

When $c = \log(C)$, this is equivalent to

$$\log(\|xT - \widehat{xT}\|_\infty) = c + \alpha \log(M) - \beta \log(\sqrt{N}) + \varepsilon, \quad \text{where, } \varepsilon \sim N(0, \sigma^2). \quad (12)$$

This formulation gives a linear model in the transformed features with homoscedastic normal residuals. Ordinary least squares (OLS) was used to estimate the unknown parameters c, α, β , and σ^2 . The fit of the estimated distribution was then assessed based on the R^2 , the normality and homoscedasticity of the residuals.

4.1.2 Acceptable model error

With the distribution of the model error known, a natural question is: what size of the model error is acceptable? A possible purpose of the Expected Threat model is to quantify the ability of a player to create threat within the setting of performance monitoring for a prospective transfer target. We used expert consultation together with simulations to determine when a model is of sufficient quality.

The values of xT created per player are dependent on the context, like playing position and the league. The xT values of a specific player should thus be interpreted relative to players in a similar situation. Therefore, a natural way of quantifying the quality of players is to consider their quartile within a group of players playing at the same position and in the same league.

In consultation with practitioners, it was established that a model error is acceptable if at most 10% of the players are assigned an incorrect quartile by the estimated

model. Additionally, the difference between the assigned quartiles of the estimated and the ground truth model can at most be one quartile. To assure this is likely to hold, these situations should be satisfied with at least 0.90 probability. In short, a model is of acceptable quality if, with probability of 0.90, less than 10% of the players are assigned the incorrect quartile with at most one quartile change.

To find to which model error this corresponds, we performed simulations. Players playing as center forwards in the French Ligue 1 in the 2015/2016 season were considered. As these players played in the same competition and position, they roughly faced the same opponents and their values can be compared. Players with less than 300 minutes as a center forward were filtered out to ensure reliable estimates. This resulted in a group of 47 players whose ratings could be compared.

A ground truth model with a 16×12 ($M = 192$) grid was used to simulate 10,000 estimated models for different values of N . Table 3 describes the different sizes of the training data set N . For all resampled models, the xT created per 90 was calculated for each player. We then determined the quartile of this player performance and compared it to the quartile assigned based on the ground truth model. For the estimated models, the model error could also be determined. This resulted in a dataset containing 130,000 samples of the model error, number of incorrectly assigned player quartiles and the largest quartile change by a player rating.

Table 3: Sample sizes N used in the simulations to find the maximal acceptable model error.

Sample sizes N		
100,000	630,000	3,000,000
130,000	1,300,000	3,500,000
170,000	1,650,000	4,000,000
240,000	2,000,000	
370,000	2,500,000	

To study the relationship between the model error and the number of players with incorrect quartile estimations, the data points were evenly divided into 75 bins with a similar model error. For each bin, the median and the 10%-90% quantile range were calculated. This was used to find the maximal acceptable size of the model error ($ME_{\max}$). This is the highest model error that had less than 10% of the players change at most one quartiles with a 0.90 probability.

4.2 Results

4.2.1 Distribution of the model error

Figure 2 shows kernel density estimates for the density of the model error based on different values of N and M before any filtering. The results show that indeed, large error sizes occur more frequently with small values of N and with a large number of game states M . This confirms that large model errors correspond to situations with a large number of game states M and a small number of training data points N .

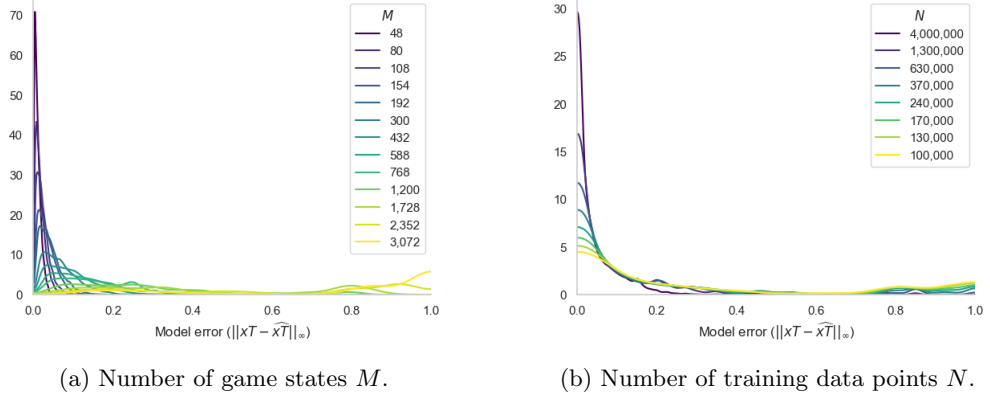


Fig. 2: The kernel density estimates of the density of the model error colored by their value of M and N

The plots generally show two modes in the distributions. One mode occurs in the vicinity of 0 and corresponds to small fluctuations of $\widehat{xT}$ around the actual values xT . The mode in the model errors between 0.8 and 1.0, on the other hand, are situations where the training data did not contain any shot in a game state with a high probability of scoring. A value of 0 is assigned to the probability of scoring directly from that state g_s , and the model greatly underestimates the actual goal-scoring probability. These situations can be observed in practice via the number of shots per state. So, the filter on $M^{3/2} \log(M)/\sqrt{N} < 445$ ensures that these situations are excluded in further analyses.

The relationship between the model error and the estimation error in the model parameters T and g is visualized in Figure 3, along with histograms of the involved quantities. The histograms indicate that errors in estimating the goal-scoring probabilities, g , are generally larger than the errors in estimating the transition matrix weighted by their xT -value. Additionally, the error in the goal-scoring probabilities has a high Pearson correlation coefficient ($r = 0.9935$) with the model error. The correlation coefficient of the weighted error in the transition matrix estimation is lower ($r = 0.8069$). The results show that the model error is not dominated by the errors in the transition matrix, but by the errors in the goal-scoring probabilities.

Table 4 shows the results of fitting the parametric model in (12) using OLS. The R^2 -value of 0.864 indicates that most of the differences between the model errors are explained by the estimated dependency on M and N . It suggests that the estimated model captures the relationship well.

Figure 4 shows the residuals plotted against the fitted values for different levels of M and N . It is visible that small values of M and large values of N have slightly more variation in the residuals. Still, this effect is relatively small. Thus, the residuals are not too far from homoscedastic.

A QQ-plot of the residuals is shown in Figure 5. The residuals roughly follow a normal distribution, although the data have a heavy left tail and a light right tail.

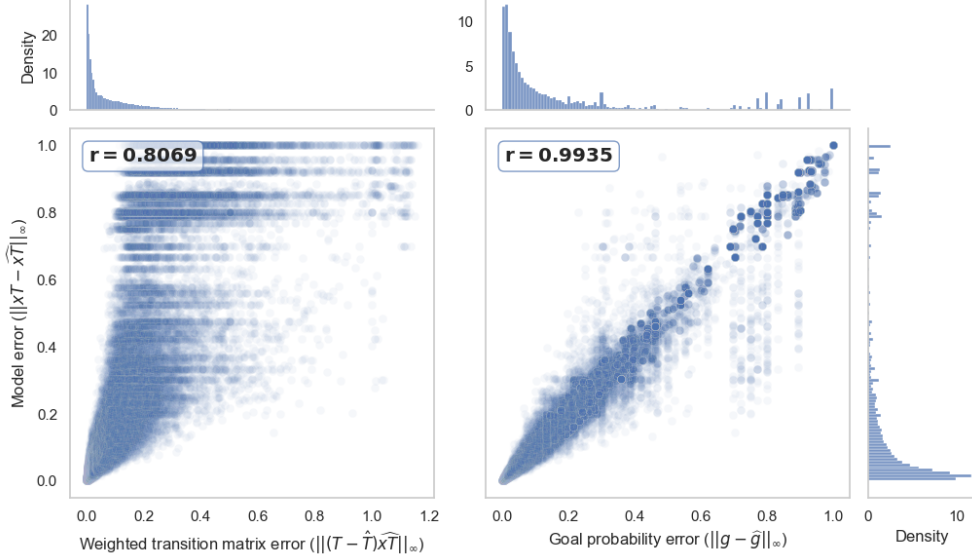


Fig. 3: A scatterplot of the error in estimating different parts of the model plotted against the model error accompanied by a histogram of the estimation errors.

This means that the model assumes that there are slightly more large errors than in reality, which translates to a small overestimation of the quantiles of the model errors in practice. As a result, the rules of thumb derived under the normality assumption will be slightly conservative, providing a margin of safety in practice.

The results indicate that the model explains most of the variance of the model error, the residuals are roughly homoscedastic and roughly normally distributed. Therefore, we conclude that the model error approximately follows a lognormal distribution on the part of the domain of M and N that is used in practice. The values of the coefficients

Dep. Variable:	$ \widehat{xT} - xT _\infty$	R-squared:	0.864			
Model:	OLS	Adj. R-squared:	0.864			
No. Observations:	76000	Log-Likelihood:	-42284.			
	coef	std err	t	P> t	[0.025	0.975]
c	-2.0916	0.017	-121.500	0.000	-2.125	-2.058
α	1.0100	0.002	635.048	0.000	1.007	1.013
β	1.0267	0.003	402.624	0.000	1.022	1.032
Variance residuals	0.1782					

Table 4: Summary of the OLS model fitted on the log model error for the data points with $M^{3/2} \log(M)/\sqrt{N} < 445$.

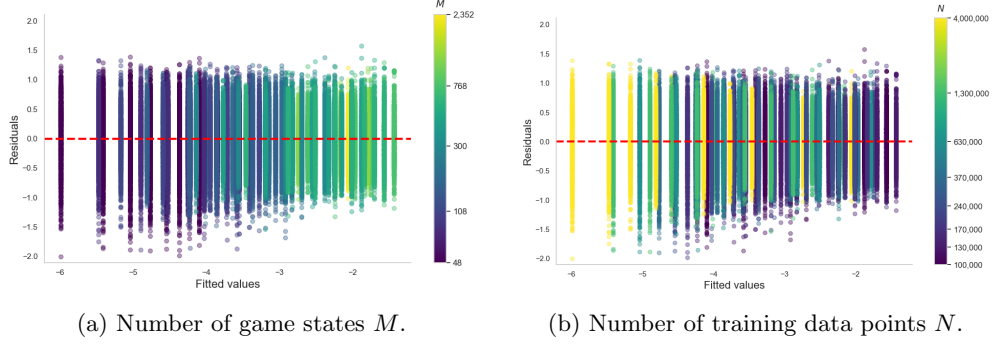


Fig. 4: The residuals versus the predicted values in the OLS model colored by their value of M and N

in Table 4 indicate that the size of the model error approximately has the following distribution.

$$||xT - \widehat{xT}||_{\infty} = e^{-2.0916} \frac{M^{1.01}}{(\sqrt{N})^{1.0267}} e^{\varepsilon}, \quad \text{where } \varepsilon \sim N(0, 0.1782) \quad (13)$$

4.2.2 Acceptable model error

Domain expert consultation indicated that a model is of acceptable quality if less than 10% of the players are assigned the incorrect quartile with at most one quartile change. This should hold with probability 0.9. The simulations indicated that more than one quartile changes rarely occurred. The relationship between the size of the model error and the number of players assigned the wrong quartile is given in Figure 6. It is visible that the number of quartile changes typically increases with the model error.

More specifically, the results show that the highest model error having less than 10% quartile changes with a probability of 0.90 is 0.0192. Therefore, it is found that the maximal acceptable model error for scouting purposes is $ME_{max} = 0.0192$.

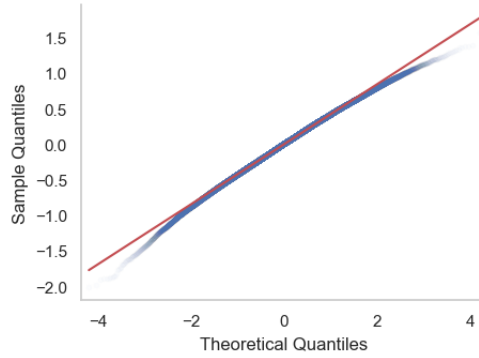


Fig. 5: A QQ-plot of the model residuals versus a standard normal distribution.

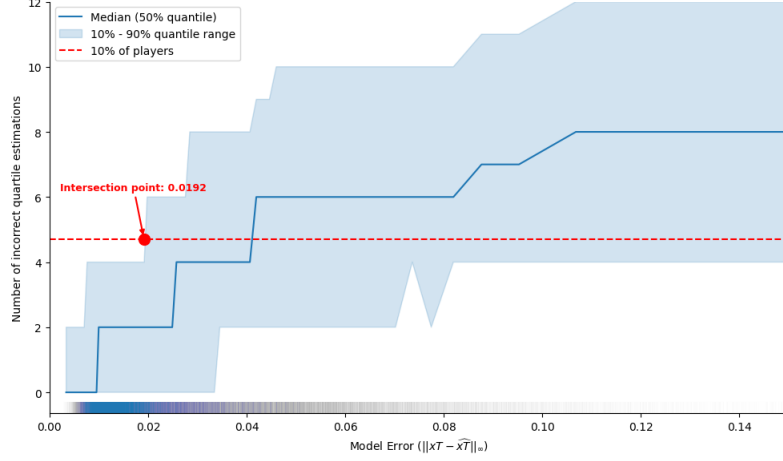


Fig. 6: The estimated median and the 10%-90% quantile range of the number of incorrect quartile assignments of players by the resampled models for different values of the model error.

5 Application

The Expected Threat model can be used to assist in decision making in real-life situations at football clubs. For instance, the model can be used to calculate the xT created per 90 minutes for different players. The xT created per 90 minutes is a performance metric that can be used to objectively compare transfer targets when trying to find a midfielder that is good at moving possession to high-threat situations.

To responsibly use the Expected Threat model, one first has to ensure that the Expected Threat model is of sufficient quality and has an acceptable model error. The findings of this study can be applied either to validate the quality of an existing model or to designing a new model of sufficient quality. For this purpose, we formulate rules of thumb based on the results in Section 4. Lastly, it is illustrated how an Expected Threat model with an acceptable model error can be used for scouting purposes on publicly available StatsBomb data of the 2020 Euros [25].

5.1 Model validity

5.1.1 Quality check of an existing model

In case football clubs already have access to an existing Expected Threat model, it is beneficial to check whether the existing model is already of sufficient quality. This prevents the unnecessary use of time and resources.

Consider the Expected Threat model by Singh [11], visualized in Figure 7b, as an example. This model was trained on a data set of the 2017-2018 Premier League season, which roughly corresponds to $N = 620,000$. It has a 16×12 grid resulting in $M = 192$. These parameters can be plugged into (13) to find that the model error behaves like a lognormal variable as visualized in Figure 7a.

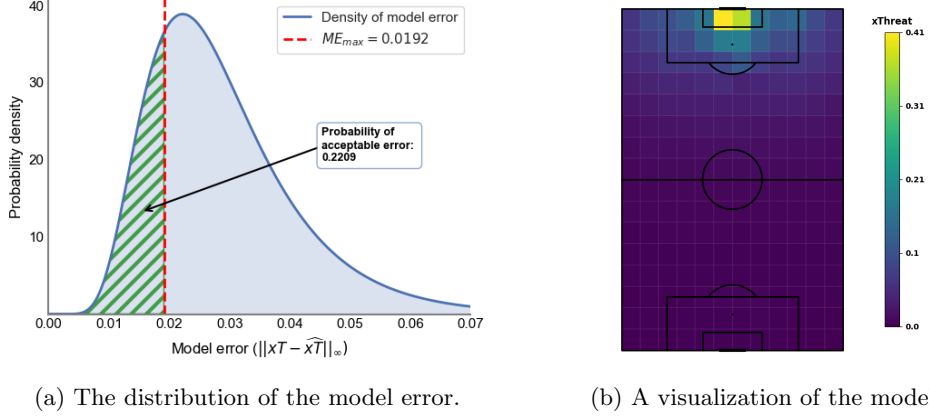


Fig. 7: The Expected Threat (xT) model introduced by [11] with $M = 192$ and $N = 620,000$.

The area under the curve with model errors smaller than $ME_{max} = 0.0192$ represents the probability of the model error being of acceptable size. Applying this criterion yields a probability of 0.2209, suggesting that the model may not meet the required threshold. Player ratings derived from this model should be interpreted with appropriate caution.

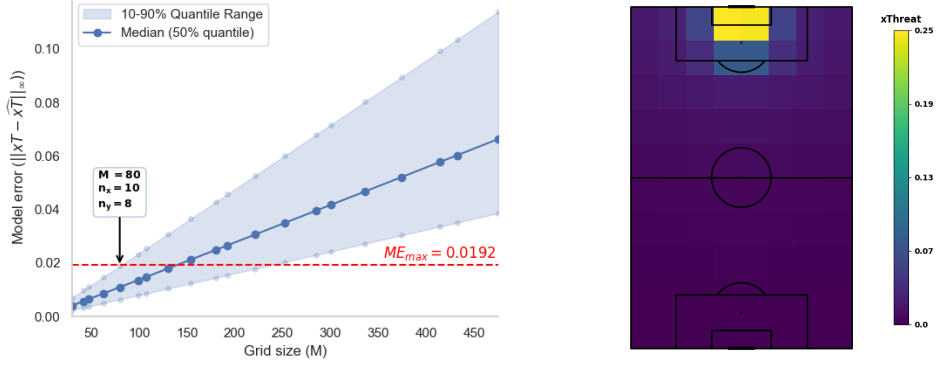
5.1.2 Selecting grid size for given training set

If no model of sufficient quality is present, it is necessary to train a new model. The most likely scenario is that there is a specific data set available to train the model on. A natural question that arises is: what is the best number of game states M given a training dataset of size N ?

When selecting the grid, there is a clear trade-off present. On the one hand, a small number of game states is desirable since it results in a smaller error in estimating the Expected Threat values. On the other hand, a model with more game states is better able to distinguish between different in-game situations, which reduces the error due to model assumptions as discussed in Section 2.4. Hence, a finer grid could be favorable.

Based on the results of the simulations in combination with expert consultation, we formulate the following rule of thumb. *Select the most flexible model (highest M) such that the model error is smaller than $ME_{max} = 0.0192$ with 0.90 probability.* This selection rule ensures that the model error is best able to distinguish between different in-game scenarios, while keeping the model error within an acceptable range.

As an illustration, we consider a situation where 4 seasons of data are available for a specific league. This corresponds to 2.4 million data points. Using that $N = 2,400,000$, Figure 8 shows the quantiles of the model error for different grid sizes and the maximal acceptable model error ME_{max} . The rule of thumb indicates to select $M = 80$, which corresponds to a 10×8 grid.



(a) The quantiles of the maximal model error of an Expected Threat model for different grid sizes. (b) A visualization of a model with the selected 10×8 grid.

Fig. 8: A visualization of the rule-of-thumb for finding the grid given $N = 2,480,000$.

5.1.3 Required data set size for a given grid

When a football club is setting up the infrastructure to use in-house data-driven modeling, it might be the case that a training data set is not yet fully present. To sufficiently distinguish between in-game situations for a specific application, the staff might desire an Expected Threat model with a specific grid and would like to know what the required size of the data set is.

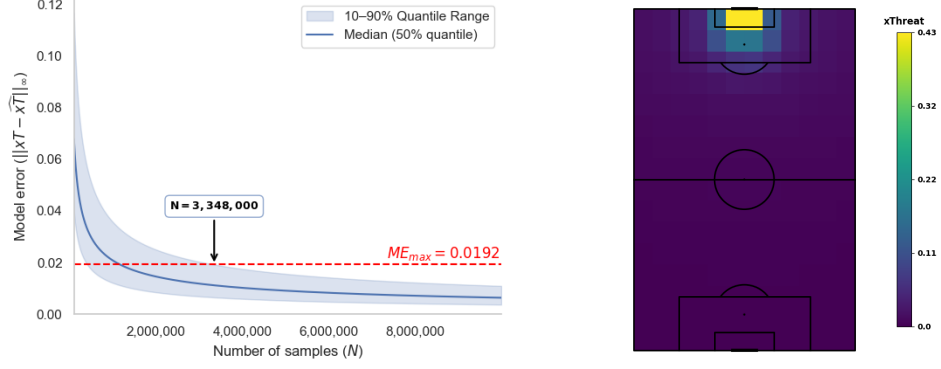
The simulations showed that a larger training data set results in a smaller model error. However, purchasing access to a larger training data set is more expensive for a football club. Additionally, one might not want to use data from too many different seasons or leagues to ensure alignment between the training data and the application. For these cases, we formulated the following rule of thumb. *Select the smallest value N such that the model error is smaller than $ME_{max} = 0.0192$ with probability 0.90.*

Assume that an Expected Threat model with a grid of 16×12 is desired. Using that $M = 192$, the quantiles of the model error for different sizes of the training set N are shown in Figure 9. The smallest value of N such that the model error is acceptable is 3,348,000. Therefore, a training set of roughly $N = 3.35$ million data points is required, corresponding to data of 5.4 seasons.

5.2 Talent identification at Euros 2020

With a validated Expected Threat model, it is possible to evaluate on-the-ball actions of players. The quality of an on-the-ball action can be determined by the increase in threat or probability of scoring a goal, as described in (1). By evaluating the quality of the in-game actions performed by a player, ratings can be obtained describing the on-the-ball quality of a player.

As an example, we use the Expected Threat model to find the midfielders who created the most goal-scoring threat at the 2020 Euros. This can, for instance, be



(a) The quantiles of the maximal model error of an Expected Threat model for different data set sizes N . (b) A visualization of a model with the selected $N = 3,348,000$.

Fig. 9: A visualization of the rule-of-thumb for finding the needed data set size given $M = 192$.

used to aid in a scouting process to identify possible targets when searching for a threat-creating midfielder.

The data used is part of the StatsBomb Open Data and is publicly available. A filter was applied that selected those who played most games as midfielder and played at least 300 minutes. For each player, ratings were obtained by calculating the total xT difference by his on-the-ball actions normalized per 90 minutes. Within this group of players, the quantiles were calculated for each player.

The results are visible in Figure 10. It shows that the xT created per 90 minutes varies between 0.055 and 0.336. Kai Havertz, Marcel Sabitzer, Mikel Damsgard, and Thomas Müller have the largest xT created and are in the fourth quartile (Q4), highlighted with dark green. Because of the relatively small number of games in the tournament, the differences cannot be interpreted as significant, especially not the small difference between Kai Havertz and Marcel Sabitzer. We conclude that these players were among the best at creating goal-scoring threat of all midfielders at the 2020 Euros.

On the other hand, Serhii Sydorchuk, Sebastian Larsson, and Albin Ekdal appear to be the worst-performing players according to this metric. However, possible biases in the data or the model should always be taken into account when evaluating the performances of players. In this specific situation, a selection bias is clearly apparent. Midfielders playing for countries that did not pass the group phase of a tournament most likely underperformed. Still, these were excluded because no reliable estimate of the xT created per 90 minutes could be made. Thus, the players not being included are likely the worst performing players, and the seemingly low-performing players in Figure 10 might have average performance. This highlights the importance of taking into account possible biases in the data.

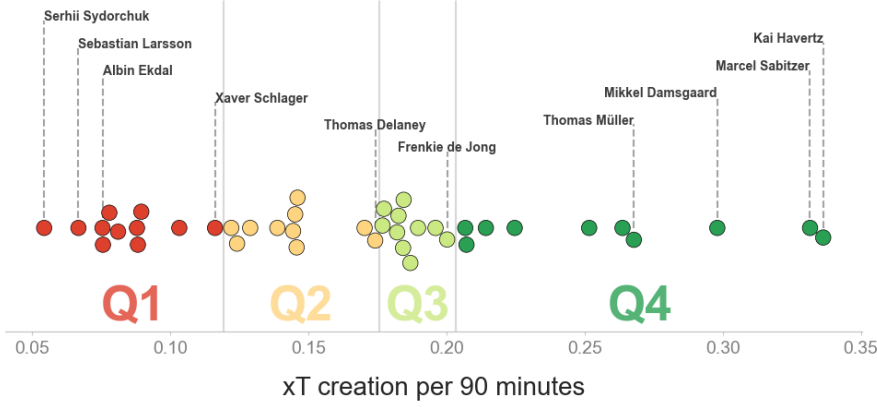


Fig. 10: The xT added per 90 minutes by midfielders with at least 300 minutes played at the 2020 Euros.

6 Discussion

The goal of this paper was (1) to gain insights into how the error of the Expected Threat model depends on the number of game states and the number of training points, and (2) to provide guidelines for practitioners to build and apply the Expected Threat model. We find that the model error behaves approximately like a lognormal random variable proportional to $M^{1.01}/N^{0.51335}$. In addition, simulations combined with expert consultations identify a maximal acceptable error of $ME_{\max} = 0.0192$. Based on this, we established practical rules of thumb to find the appropriate number of game states (M) or the desired number of training points (N) when constructing a model. An application to the 2020 Euros further illustrates how player performance can be evaluated relative to comparable players and with awareness of potential biases.

The theoretical analysis yields a worst-case bound on the model error of order $M^{3/2} \log(M)/\sqrt{N}$. Simulations showed that this bound is overly pessimistic. The model error is mostly determined by the errors in the goal-scoring probabilities, which are of a lower order than the errors in the estimated transition matrix. Consequently, the model error behaves approximately lognormal like

$$\|xT - \widehat{xT}\|_{\infty} = e^{-2.0916} \frac{M^{1.01}}{(\sqrt{N})^{1.0267}} e^{\varepsilon}, \quad \text{where } \varepsilon \sim N(0, 0.1782).$$

We conclude that the typical model performance is substantially better than the worst-case theoretical bound.

The model error seems to be largely influenced by the estimation of the direct goal-scoring probabilities. High-threat states are rarely observed, because defending teams actively prevent ball possession there. With many game states M and few data points N , a lack of observed shots can cause the model to assign zero probability of scoring directly in these high-value states, causing large errors. Consequently, the most fruitful

improvements to the Expected Threat model could be attained by enhancing the estimation of the goal-scoring probabilities. Leveraging existing shooting probabilities and xG values to estimate the corresponding quantities estimations will lead to more efficient estimators. Enforcing symmetry, applying smoothing techniques or incorporating prior knowledge could also boost the performance based on domain knowledge.

The maximal acceptable error ($ME_{\max}$) of 0.0192 means that the absolute error of the xT for each game state must be smaller than 1.92% for applications in player scouting. Existing models, such as that of Singh [11], might fall short of this threshold. Notably, the rules of thumb suggests that approximately 5.4 seasons of data are required for a 16×12 grid, far more than typically used in practice. Model validation methods such as visual inspection by domain experts and verification methods such as described in [26] can be performed to complement the rules of thumb described. Sufficient technical expertise is needed to perform this. Under these circumstances, less conservative values might be considered instead of $ME_{\max} = 0.0192$. This highlights the need for caution when applying the Expected Threat or related possession value models.

In the simulations, models trained on an open-access data set of roughly 4 million data points were assumed as the ground truth, which constitutes a limitation. These models can contain estimation errors and may be less smooth than the true underlying values. This might make the estimation task more difficult, and consequently, both the model error and the maximal acceptable error may be slightly overestimated. Future research with larger, probably proprietary, data sets could evaluate the exact impact of this effect.

Within the broader set of Expected Possession Value models [17–20], there exist other models that incorporate additional in-game context. Like the Expected Threat model, these EPV models implicitly rely on the Markov assumption. The simulations and rules of thumb in this paper can be extended in a similar manner to such models as well, enabling a systematic assessment of model quality. In this way, this paper provides a framework to quantify model quality, despite the ground truth being an intangible object in football analytics.

Appendix A Proofs of propositions and theorems

The propositions and theorems in this paper were formulated with the ℓ_∞ -norm. Most of them also hold for general vector norms with corresponding vector-induced matrix norms. The proofs below do not specify the norm if they hold in this setting.

Proof of Proposition 1 The k 'th iteration of Algorithm 1 can alternatively be formulated as

$$\widehat{xT}^{(k)} = \sum_{\ell=0}^{k-1} T^\ell g. \quad (\text{A1})$$

Because $\|\hat{T}\| < 1$, a $\widehat{xT}$ can similarly be formulated as

$$\widehat{xT} = \sum_{\ell=0}^{\infty} T^\ell g. \quad (\text{A2})$$

With these expressions, it follows that

$$\widehat{xT} - \widehat{xT}^{(k)} = \sum_{\ell=k}^{\infty} \hat{T}^\ell \hat{g} = \hat{T}^k \sum_{\ell=0}^{\infty} \hat{T}^\ell \hat{g}.$$

Since $\|\hat{T}\| < 1$, the geometric series give that

$$\|\widehat{xT} - \widehat{xT}^{(k)}\| \leq \|\hat{T}\|^k \sum_{\ell=0}^{\infty} \|\hat{T}\|^\ell \|\hat{g}\| = \frac{\|\hat{g}\| \|\hat{T}\|^k}{1 - \|\hat{T}\|}.$$

□

Proof of Proposition 2 By definition, (3) holds for both xT and $\widehat{xT}$. Consequently,

$$\begin{aligned} \|xT - \widehat{xT}\| &\leq \|g - \hat{g}\| + \|T \cdot xT - \hat{T} \cdot \widehat{xT}\| \\ &\leq \|g - \hat{g}\| + \|T\| \|xT - \widehat{xT}\| + \|(T - \hat{T})\widehat{xT}\|. \end{aligned}$$

Because $\|T\| < 1$, this is equivalent to (6). □

Proof of Proposition 3 Denote $\varepsilon = \sqrt{\frac{\log(2M/\alpha)}{2\bar{N}_s}}$. The Hoeffding's inequality states that

$$P(|g - \hat{g}_i| \geq \varepsilon) \leq 2 \exp(-2\bar{N}_s \varepsilon^2).$$

Consequently,

$$\begin{aligned} P(\|g - \hat{g}\|_\infty \geq \varepsilon) &= P\left(\max_{i=1, \dots, M} |g_i - \hat{g}_i| \geq \varepsilon\right) \\ &= P\left(\bigcup_{i=1, \dots, M} \{|g_i - \hat{g}_i| \geq \varepsilon\}\right) \\ &\leq \sum_{i=1}^M P(|g_i - \hat{g}_i| \geq \varepsilon) \\ &\leq 2M \exp(-2\bar{N}_g \varepsilon^2) \\ &= \alpha. \end{aligned}$$

Consequently, the inequality holds with probability $1 - \alpha$. The equality of the bounds follows from the definition of $\bar{N}_g$. □

Proof of Proposition 4 Denote $\varepsilon = M\sqrt{\frac{\log(2M^2/\alpha)}{2\bar{N}_m}}$. Note that Hoeffding's inequality gives

$$P(|T_{i,j} - \hat{T}_{i,j}| \geq \varepsilon/M) \leq 2\exp(-2\bar{N}_m\varepsilon^2/M^2)$$

Then,

$$\begin{aligned} P(\|T - \hat{T}\|_\infty \geq \varepsilon) &\leq P\left(M \max_{i,j=1,\dots,M} |T_{i,j} - \hat{T}_{i,j}| \geq \varepsilon\right) \\ &= P\left(\bigcup_{i,j=1,\dots,M} M|T_{i,j} - \hat{T}_{i,j}| \geq \varepsilon\right) \\ &\leq \sum_{i,j=1,\dots,M} P(|T_{i,j} - \hat{T}_{i,j}| \geq \varepsilon/M) \\ &\leq \sum_{i,j=1,\dots,M} 2\exp(-2\bar{N}_m\varepsilon^2/M^2) \\ &= 2M^2 \exp(-2\bar{N}_m\varepsilon^2/M^2) \\ &= \alpha. \end{aligned}$$

Consequently, the inequality holds with probability $1 - \alpha$. The equality of the bounds follows from the definition of $\bar{N}_m$. $\square$

Proof of Theorem 1 The inequality follows directly from combining Proposition 1, Proposition 2, Proposition 3, and Proposition 4. Hereby, it is used that $\|(T - \hat{T})x\|_\infty \leq \|T - \hat{T}\|_\infty$. $\square$

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